

Index of Industrial Production (IIP) - Compilation Methodology

GOVERNMENT OF INDIA

MINISTRY OF STATISTICS AND PROGRAMME IMPLEMENTATION (MoSPI)

NATIONAL STATISTICAL OFFICE (NSO) - ECONOMIC STATISTICS DIVISION (ESD)

COMPILATION METHODOLOGY: INDEX OF INDUSTRIAL PRODUCTION (IIP) - BASE 2011-12=100

CHAPTER 2: ITEM BASKET, WEIGHTING DIAGRAM AND LASPEYRES AGGREGATION

1. Objective and Scope of IIP

The Index of Industrial Production (IIP) is an abstract number indicating the salient volume changes in the production of an industry basket compared to a reference base period.

In India, the current series of IIP uses 2011-12 as the base year and measures growth in three broad sectors:

- Mining (Sector Weight: 14.37%)
- Manufacturing (Sector Weight: 77.63%)
- Electricity (Sector Weight: 7.99%)

2. Use-Based Classification

Beyond sectoral compilation, IIP provides policy insights through a 6-tier Use-Based Classification:

- 1. Primary Goods (Weight: 34.05%):** Mining, electricity, fuels, basic fertilizers.
- 2. Capital Goods (Weight: 8.22%):** Machinery, commercial vehicles, boilers, transformers used in gross fixed capital formation.
- 3. Intermediate Goods (Weight: 17.22%):** Yarns, chemical intermediates, steel rods, paints used as inputs in subsequent manufacturing.
- 4. Infrastructure / Construction Goods (Weight: 12.34%):** Cement, tiles, structural pipes, fixtures.
- 5. Consumer Durables (Weight: 12.84%):** Refrigerators, passenger vehicles, consumer electronics, furniture.
- 6. Consumer Non-Durables (Weight: 15.33%):** Packaged food, pharmaceuticals, soaps, edible oils, apparel.

3. Source Data Collection Agencies

Monthly production data are received from 14 source agencies across administrative ministries:

- Department for Promotion of Industry and Internal Trade (DPIIT) - Major source for manufacturing.
- Indian Bureau of Mines (IBM) - Mining production figures.
- Central Electricity Authority (CEA) - Electricity generation from thermal, hydro, nuclear, and renewable sources.
- Directorate of Sugar and Vegetable Oils, Coffee Board, Tea Board, Office of the Textile Commissioner.

4. Aggregation Formula

The index compilation adheres strictly to Laspeyres formula with base period weights:

$$IIP_t = \frac{\sum (W_i \times \frac{R_{it}}{R_{i0}})}{\sum W_i} \times 100$$

Where:

- IIP_t is the composite index for month t .
- W_i is the assigned weight of item i in the base period 2011-12.
- R_{it} is the reported physical production quantity of item i in month t .
- R_{i0} is the base monthly average production of item i in 2011-12.

5. Treatment of Non-Response and Revisions

- Quick Estimates (QE): Released on the 12th of every month (or previous working day) with a time lag of six weeks.
- First Revision: Carried out one month later alongside the subsequent release.
- Final Revision: Compiled three months after initial release when late production returns from factories are validated.

Non-response is handled using item-specific imputation based on the growth rates of reporting units within the same 4-digit NIC class.